

Here is the complete setup for **Program 03** and **Program 04** with both the **XML layout** code and the clean **Kotlin** source code, matching the exact UI design requirements from your lab manual.

Program 03: Developing a Login Window Using UI Controls

This program creates a login screen with fields for a username and password. The Kotlin code validates the input and shows a Toast message based on whether the login succeeded or failed.

1. Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="100dp"
        android:text="USER NAME"
        android:textSize="24sp"
        android:textStyle="bold" />

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@id/textView"
        android:layout_marginTop="16dp"
        android:hint="Enter the Name"
        android:ems="10"
        android:inputType="text"
        android:minHeight="48dp" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```

        android:layout_below="@id/editTextText"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="32dp"
        android:text="PASSWORD"
        android:textSize="24sp"
        android:textStyle="bold" />

<EditText
    android:id="@+id/editTextText2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/textView2"
    android:layout_marginTop="16dp"
    android:hint="Enter the Password"
    android:ems="10"
    android:inputType="textPassword"
    android:minHeight="48dp" />

<Button
    android:id="@+id/button"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/editTextText2"
    android:layout_marginTop="40dp"
    android:text="LOGIN"
    android:textSize="20sp"
    android:textStyle="bold" />

</RelativeLayout>

```

2. Activity (MainActivity.kt)

```

package com.example.loginapp

import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState) {
        super.onCreate(savedInstanceState)

        [span_6] (start_span) setContentView(R.layout.activity_main) [span_6] (end
        _span)
    }
}

```

```

        [span_7](start_span)// Bind UI views using the layout
IDs[span_7](end_span)
        val usernameEditText =
findViewById<EditText>(R.id.editTextText)
        val passwordEditText =
findViewById<EditText>(R.id.editTextText2)
        val loginButton = findViewById<Button>(R.id.button)

        [span_8](start_span)// Set up the click action for the login
button[span_8](end_span)
        loginButton.setOnClickListener {
            [span_9](start_span)val username =
usernameEditText.text.toString().trim()[span_9](end_span)
            [span_10](start_span)val password =
passwordEditText.text.toString().trim()[span_10](end_span)

            [span_11](start_span)[span_12](start_span)// Credentials
check logic[span_11](end_span)[span_12](end_span)
            [span_13](start_span)if (username.isEmpty() ||
password.isEmpty()) {[span_13](end_span)
                [span_14](start_span)Toast.makeText(this, "Username or
Password cannot be empty",
Toast.LENGTH_SHORT).show()[span_14](end_span)
                [span_15](start_span)} else if (username == "admin" &&
password == "pass") {[span_15](end_span)
                    [span_16](start_span)Toast.makeText(this, "Login
Successful", Toast.LENGTH_SHORT).show()[span_16](end_span)
                } else {
                    [span_17](start_span)Toast.makeText(this, "login
fail", Toast.LENGTH_SHORT).show()[span_17](end_span)
                }
            }
        }
    }
}

```

Program 04: Implement New Activity Using Explicit and Implicit Intent

This application uses an **Implicit Intent** to open a web page in a browser, and an **Explicit Intent** to transition to a secondary screen within the app.

1. Dashboard Layout (activity_main.xml)

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout

```

```

xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/button"
        android:layout_width="220dp"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="250dp"
        android:text="Implicit Intent" />

    <Button
        android:id="@+id/button2"
        android:layout_width="220dp"
        android:layout_height="wrap_content"
        android:layout_below="@id/button"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="24dp"
        android:text="Explicit Intent" />

</RelativeLayout>

```

2. Second Screen Layout (activity_new.xml)

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    tools:context=".SecondActivity">

    <TextView
        android:id="@+id/tv"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20dp"
        [span_21] (start_span) android:text="Welcome to explicit intent"
    />[span_21] (end_span)

</RelativeLayout>

```

3. Dashboard Activity (MainActivity.kt)

```
package com.example.intentdemo

import android.content.Intent
import android.net.Uri
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState) {
        super.onCreate(savedInstanceState)

        [span_22] (start_span) setContentView(R.layout.activity_main) [span_22] (end_span)

        val implicitBtn = findViewById<Button>(R.id.button)
        val explicitBtn = findViewById<Button>(R.id.button2)

        [span_23] (start_span) // Implicit Intent: Requests the system
        open a web link [span_23] (end_span)
        implicitBtn.setOnClickListener {
            [span_24] (start_span) val webUri =
            Uri.parse("https://www.google.com") [span_24] (end_span)
            [span_25] (start_span) val intent =
            Intent(Intent.ACTION_VIEW, webUri) [span_25] (end_span)

            [span_26] (start_span) startActivity(intent) [span_26] (end_span)
        }

        [span_27] (start_span) // Explicit Intent: Directly calls the
        target class within this application [span_27] (end_span)
        explicitBtn.setOnClickListener {
            [span_28] (start_span) val intent = Intent(this,
            SecondActivity::class.java) [span_28] (end_span)

            [span_29] (start_span) startActivity(intent) [span_29] (end_span)
        }
    }
}
```

4. Second Activity Screen (SecondActivity.kt)

```
package com.example.intentdemo

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity

class SecondActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState) {
        super.onCreate(savedInstanceState)

        [span_30] (start_span) setContentView(R.layout.activity_new) [span_30] (en
d_span)
    }
}
```